



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 2 Structured Questions (Solutions)

9729/02
12 September 2018
2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer all questions in the spaces provided on the Question Paper.
The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

For Examiner's Use	
1	/9
2	/9
3	/22
4	/13
5	/13
6	/9
TOTAL	/ 75

1	Elements in Period 3 exhibit a variety of physical and chemical properties.	
	(a)	An element in Period 3 has a high melting point but low electrical conductivity. Identify this element and explain why it has a high melting point using concepts of structure and bonding. [1]
		Silicon has a giant molecular structure and strong covalent bonds between the atoms in the molecule . Hence it needs a large amount of energy to overcome it. [1]
	(b)	Two chlorides of Period 3 elements exist as liquids at room temperature. These two chlorides react with water to give white fumes, but only one of them gives a white solid.
	(i)	Using concepts of structure and bonding, explain why these two chlorides have low boiling points and exist as liquid at room temperature. [1]
		Both have simple molecular structure . These compounds have low boiling points since less heat energy is required to overcome the weak intermolecular forces of attraction . [1]
	(ii)	Write chemical equations, with state symbols, for each of these chlorides reacting with water. [2]
		$\text{SiCl}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{SiO}_2(\text{s}) + 4\text{HCl}(\text{g})$ [1] $\text{PCl}_3(\text{l}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_3(\text{aq}) + 3\text{HCl}(\text{g})$ [1] Do not accept PCl_5 as it is a solid
	(c)	Period 3 elements also form oxides which react with water. Na_2O and SO_2 are two such oxides.
	(i)	Write chemical equations, with state symbols, when each of these oxides react with water. [2]
		$\text{Na}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq})$ [1] $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{SO}_3(\text{aq})$ [1]
	(ii)	SO_2 is used as a food preservative. Suggest the property of SO_2 which enables it to function this way. [1]
		It acts as a reducing agent/ antioxidant. [1] Note: Food may turn bad due to oxidation process. SO_2 will be oxidised in place of the food.
	(d)	Sulfur forms the compound S_4N_4 with nitrogen. The structure of S_4N_4 is shown below. Assume all bonds shown are single bonds.

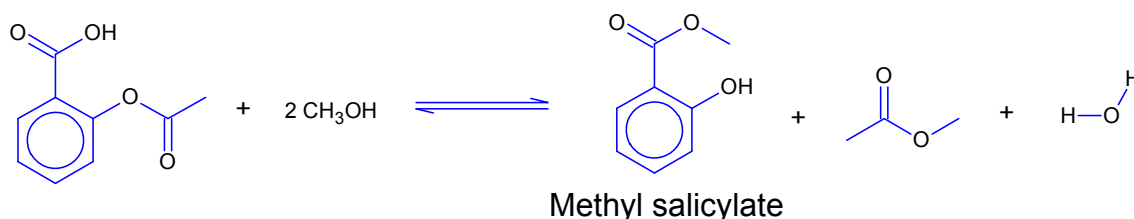
		Determine the number of lone pairs of electrons around a sulfur atom and a nitrogen atom in S ₄ N ₄ .	
	(i)	Nitrogen atom Sulfur atom	[1]
		Nitrogen atom: <u>1</u> lone pair Sulfur atom: <u>2</u> lone pairs [1]	
	(ii)	Which bond angle, a or b , will be smaller? Explain your answer.	[1]
		Bond angle a will be smaller . Lone-pair lone-pair repulsion is greater than lone-pair single electron repulsion. [1]	
			[Total: 9]

2	The following graph shows the change in mass with time for the decomposition of 10.0 g of magnesium carbonate and 10.0 g of strontium carbonate under the same conditions:		
(a)	(i)	Calculate the value of y .	[2]
		$\text{SrCO}_3(\text{s}) \rightarrow \text{SrO}(\text{s}) + \text{CO}_2(\text{g})$ y is the mass of SrO that formed from the decomposition $\text{Amount of SrCO}_3 = \frac{10}{(87.6 + 12.0 + 16.0 \times 3)} = 0.06775 \text{ mol [1]}$ $\text{Amount of SrO formed} = 0.06775 \text{ mol}$ $\text{Mass of SrO, } y = 0.06775 \times (87.6 + 16.0) = 7.02 \text{ g [1]}$	

	(ii)	Explain why the value of t_2 is larger than t_1 . [3]												
		• <u>Cationic radius of Sr^{2+} is larger than Mg^{2+} while cationic charge is the same or Charge density, and hence polarising power, of the Sr^{2+} is smaller.</u> [1] • <u>Polarising effect on the anion by Sr^{2+} is smaller</u> [1] and the anion electron cloud is distorted to a smaller extent. • More energy is required to overcome the C–O bond in SrCO_3. • <u>Rate is slower and a longer time (t_2) is needed to decompose SrCO_3.</u> [1]												
	(b)	There are three bottles labelled A, B and C in the laboratory. Each bottle contains one of the following reagents: aqueous Cl_2, KI solution and KBr solution. The following tests were carried out and the results were summarised in the table below. <table border="1"> <thead> <tr> <th>Experiment</th><th>Procedure</th><th>Observations</th></tr> </thead> <tbody> <tr> <td>1</td><td>mixing reagent in bottle A with reagent in bottle B</td><td>mixture remains colourless</td></tr> <tr> <td>2</td><td>mixing reagent in bottle A with reagent in bottle C</td><td>mixture turns brown</td></tr> <tr> <td>3</td><td>mixing reagent in bottle B with reagent in bottle C</td><td>mixture turns brown</td></tr> </tbody> </table>	Experiment	Procedure	Observations	1	mixing reagent in bottle A with reagent in bottle B	mixture remains colourless	2	mixing reagent in bottle A with reagent in bottle C	mixture turns brown	3	mixing reagent in bottle B with reagent in bottle C	mixture turns brown
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	(i)	Which bottle contains aqueous Cl_2 ? With the aid of a balanced equation, explain your answer. [2]												
		Bottle C [1] $\text{Cl}_2 + 2\text{X}^- \longrightarrow 2\text{Cl}^- + \text{X}_2$ where $\text{X} = \text{Br}$ or I [1] Down the group, reduction potential becomes less positive. The halogens have lower tendency to be reduced. <u>Stronger oxidising agent higher in the Group oxidises</u> (and hence displaces) the <u>halide ions in aqueous solution further down the Group</u> [1]												
	(ii)	If hexane is also provided, how would you use it to identify the contents of the other two bottles? Include the observations in your answer. [1]												
		Knowing that bottle A and B is either KBr or KI, <u>add hexane to the two brown mixtures obtained, separately.</u> If the organic layer is <u>purple, bottle contains KI.</u> If the organic layer is <u>red-brown, bottle contains KBr.</u> [1]												

	(c)	Using relevant data from the <i>Data Booklet</i> , comment on the thermal stability of hydrogen bromide and hydrogen chloride.	[1]
		Bond energy of H-Br (366 kJ mol^{-1}) is lesser than bond energy of H-Cl (431 kJ mol^{-1}) Less energy is required to overcome the weaker covalent bond in H-Br thus HBr will decompose at a lower temperature. [1]	
			[Total:9]

- 3** Methyl salicylate, commonly known as oil of Wintergreen is used as a flavouring agent in candy. When methyl salicylate is applied to the skin, it causes a mild burning sensation which serves as a counter-irritant for sore muscles. It can be synthesise from aspirin. The reaction is as follows.



Reagent	Density / g cm ⁻³	M _r	Solubility in water
Acetylsalicylate acid (Aspirin)		180.0	insoluble
Sulfuric acid	1.84	98.0	soluble
Methanol	0.792	32.0	soluble
Methyl salicylate	1.17	152.1	insoluble
Water	1.00	18.0	

Preparation of impure methyl salicylate

1. Weigh approximately 20 g of acetylsalicylate acid into a round-bottom flask. Add 30 cm³ of methanol and stir until all the acetylsalicylate acid has dissolved.
2. Place 25 cm³ of concentrated sulfuric acid in the tap funnel and then add the acid dropwise into the reagents in the flask. Keep the contents well shaken and cool occasionally in an ice-water bath.
3. When all of the acid have been added, replace the tap funnel with a reflux condenser and gently boil the mixture for an hour.
4. Remove the condenser and boil off about 50% of the volume.

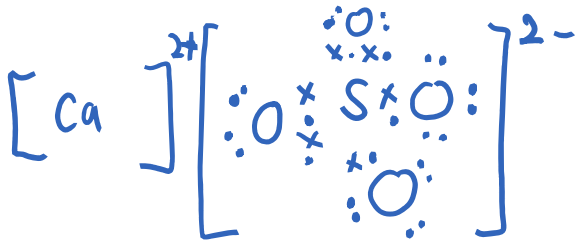
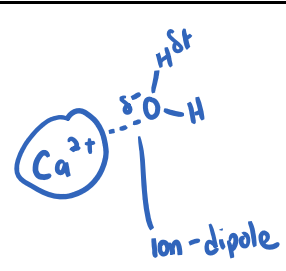
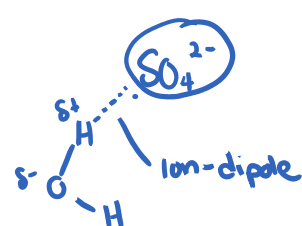
- (a)** **(i)** By using the amounts given above, prove that methanol is in excess and suggest why it is done to increase yield.

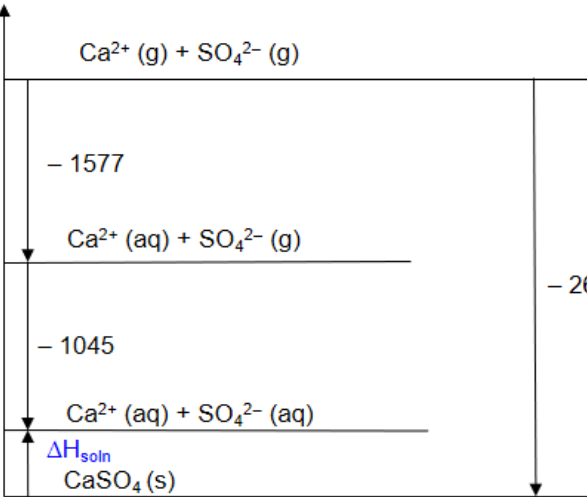
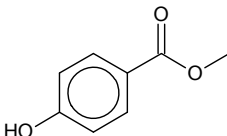
[2]

$n_{\text{methanol}} = (30 \times 0.792) / 32 = \underline{0.742 \text{ mol}}$
 $n_{\text{aspirin}} = 20/180 = \underline{0.1111 \text{ mol}}$
 $n_{\text{methanol}} = \frac{1}{2} \text{ aspirin}$
 Hence methanol is in excess. [1]

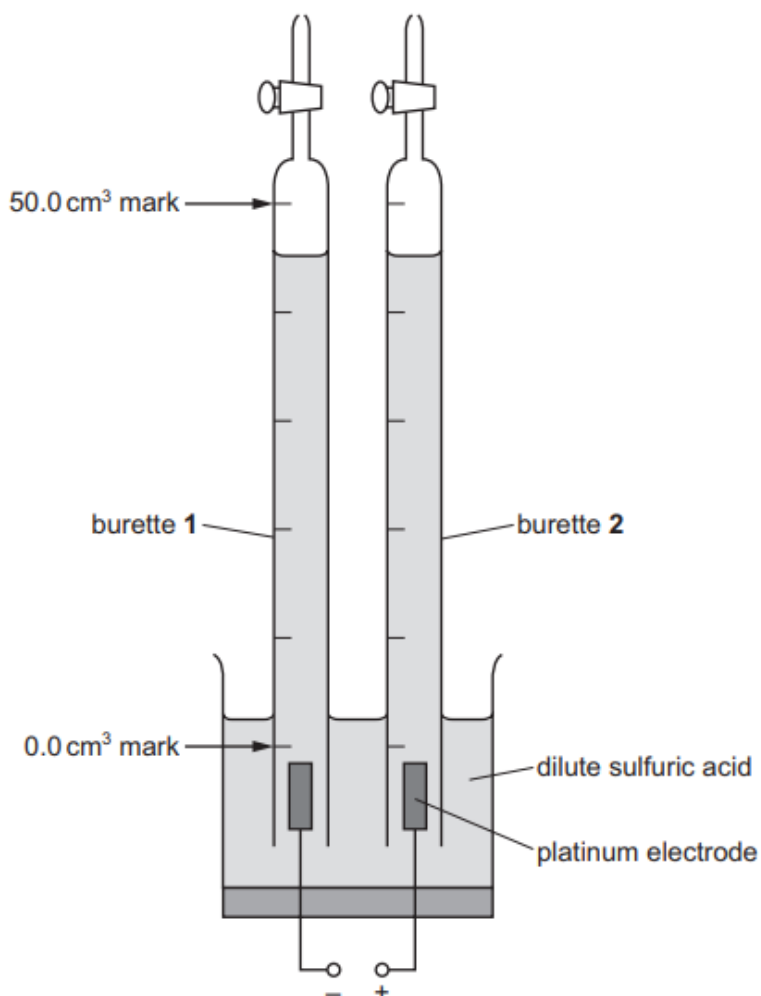
		By LCP, increase the methanol concentration would cause <u>the position of equilibrium to lie towards the forward reaction (right)</u> , [1] increasing the amount of methyl salicylate produced.
(b)		When concentrated sulfuric acid is added to the reaction mixture, cooling is necessary. Suggest the main cause of heat being produced at this stage. [1]
		<u>Dilution of concentrated sulfuric acid produces heat.</u> [1]
(c)		The reaction mixture was heated overnight. Why is this process necessary for the preparation of many covalent organic compounds? [1]
		<u>Breaking strong covalent bonds</u> [1] requires a large amount of energy to overcome.
		The crude product formed requires purification as it contains many impurities. Purification of impure methyl salicylate 5. Transfer the remaining mixture into a separatory funnel. Shake the mixture with 30 cm³ of ice cool water and separate the aqueous layer from the methyl salicylate. Reject the aqueous layer. 6. Return the methyl salicylate to the funnel. Add 20 cm³ of dilute sodium hydrogencarbonate in the separating funnel, inverting the flask and opening the tap at intervals to prevent build up of pressure. 7. Transfer the methyl salicylate into a conical flask and add some granular anhydrous calcium sulfate. Swirl the mixture until the liquid is clear. 6. Filter the methyl salicylate into a clean vial. Weigh the purified
(d)		Volume of methyl salicylate produced is 10 cm ³ . Calculate the percentage yield.
		<u>Aspirin is the limiting reaction</u> $n_{\text{methyl salicylate}} = n_{\text{aspirin}} = 20/180 = \underline{0.1111\text{mol}}$ [1] Theoretical yield (in Vol) = $0.1111 \times 152.1 \div 1.17 = 14.44 \text{ cm}^3$ % yield = $10/14.44 = \underline{69.2 \%}$ [1]
(e)		The impure methyl salicylate was shaken with water (step 5) and the two layers are allowed to separate.

	(i)	By considering the structure and bonding of methyl salicylate, suggest why it is immiscible with methanol. [2]
		The <u>extensive intermolecular id-id interactions</u> due to the <u>hydrophobic benzene ring</u> [1] is <u>not strong enough</u> to <u>displace the stronger intermolecular hydrogen bonds</u> . [1]
	(ii)	Suggest the organic compounds inside both layers. Explain your answer. [2]
		Top layer: <u>methanol</u> Bottom layer: <u>methyl salicylate</u> [1] for both Explanation: <u>methyl salicylate is more dense</u> than methanol [1] for reason

(f)	(i)	It is important to open the tap at intervals to prevent build up of pressure in when dilute aqueous sodium hydrogencarbonate was added (step 6). Explain, with the aid of an equation why this is important. [2]
		$\text{H}_2\text{SO}_4 + 2\text{NaHCO}_3 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O} + \text{Na}_2\text{SO}_4$ [1] (x) but mark reason separately – $\text{H}_2\text{SO}_4 + \text{NaHCO}_3 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{NaHSO}_4$ $\text{H}_2\text{SO}_4 + \text{methyl salicylate} \rightarrow \text{CO}_2$ <u>Gaseous CO₂</u> is produced which will <u>increase the pressure</u> in the separatory funnel. [1]
(g)	(i)	Draw the dot-and-cross diagram of calcium sulfate used in step 7. [2]
		 [1] [1]
	(ii)	Calcium sulfate is a desiccant to remove any water left in the methyl salicylate product. When the desiccant has dissolved, the anions and cations are each surrounded by a number of water molecules. Draw simple diagrams to show how a water molecule can be attached to a calcium cation, and to a sulfate anion. Label each of your diagram to show the type of interaction involved. [2]
		 Ca²⁺ cation [1]  SO₄²⁻ anion [1]

	(iii)	The lattice energy of calcium sulfate is $-2640 \text{ kJ mol}^{-1}$. Use the following values of enthalpy change of hydration to construct an energy cycle and use it to calculate the enthalpy change of solution, $\Delta H_{\text{sol}}^\ominus$, for calcium sulfate. <table border="1"><thead><tr><th></th><th>$\Delta H_{\text{hyd}}^\ominus / \text{kJ mol}^{-1}$</th></tr></thead><tbody><tr><td>Ca^{2+}</td><td>-1577</td></tr><tr><td>SO_4^{2-}</td><td>-1045</td></tr></tbody></table>		$\Delta H_{\text{hyd}}^\ominus / \text{kJ mol}^{-1}$	Ca^{2+}	-1577	SO_4^{2-}	-1045
	$\Delta H_{\text{hyd}}^\ominus / \text{kJ mol}^{-1}$							
Ca^{2+}	-1577							
SO_4^{2-}	-1045							
		[3] kJ mol⁻¹  [2] energy level $\Delta H_{\text{sol}}^\ominus = \Delta H_{\text{hyd}}^\ominus (\text{Ca}^{2+}) + \Delta H_{\text{hyd}}^\ominus (\text{SO}_4^{2-}) - \Delta H_{\text{latt}}^\ominus$$= -1577 + (-1045) - (-2640) = +18.0 \text{ kJ mol}^{-1} \text{ [1]}$						
	(h)	Methyl 4-hydroxybenzoate is an isomer of methyl salicylate.  methyl 4-hydroxybenzoate						
		Predict and explain whether methyl 4-hydroxybenzoate or methyl salicylate would have a higher boiling point. [3]						
		Both have simple molecular structure. <u>Methyl 4-hydroxybenzoate would have a higher boiling point [1] than methyl salicylate.</u> The <u>close proximity</u> of the two substituents on methyl salicylate allows for <u>intramolecular hydrogen bonding</u>[1] as compared to methyl 4-hydroxybenzoate which is far apart. There will be <u>less energy</u> required to overcome the <u>less extensive intermolecular hydrogen bonds</u> [1] to overcome in methyl salicylate and thus it has a lower boiling point. [Total: 22]						

- 4 Dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, can be electrolysed using platinum electrode and a direct current. Hydrogen gas is produced at the cathode and oxygen gas is formed at the anode. The two gases are collected separately in burettes filled with dilute sulfuric acid placed over each electrode.



Reaction at electrode in burette 1: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$

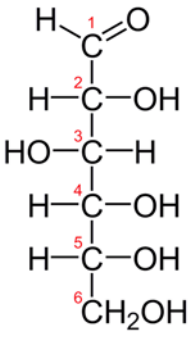
Reaction at electrode in burette 2: $\text{H}_2\text{O}(\text{l}) \rightarrow \frac{1}{2} \text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$

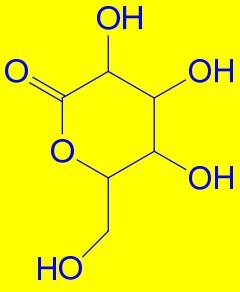
The production of hydrogen gas over time can be measured, and the data used to determine charge of one mole of electrons, known as the Faraday constant, F .

(a)	(i)	Student A performed the experiment and the volume of hydrogen gas produced during the electrolysis process were recorded in the table. Process the results to calculate the volume of the hydrogen gas produced, in cm^3, and the charge passed, in coulombs, C. The current was kept constant at 0.80A. <table><thead><tr><th>Time/s</th><th>Reading on burette cm^3</th><th>Volume of hydrogen gas produced cm^3</th><th>Charge passed /C</th></tr></thead><tbody><tr><td>0</td><td>46.20</td><td>0.00</td><td>0</td></tr><tr><td>50</td><td>41.20</td><td>5.00</td><td>40</td></tr><tr><td>100</td><td>36.20</td><td>10.00</td><td>80</td></tr><tr><td>150</td><td>31.45</td><td>14.75</td><td>120</td></tr><tr><td>200</td><td>25.80</td><td>20.40</td><td>160</td></tr><tr><td>250</td><td>20.80</td><td>25.40</td><td>200</td></tr></tbody></table> [2] [1]: Volume of H_2 correct and to 2 d.p [1] charge correct	Time/s	Reading on burette cm^3	Volume of hydrogen gas produced cm^3	Charge passed /C	0	46.20	0.00	0	50	41.20	5.00	40	100	36.20	10.00	80	150	31.45	14.75	120	200	25.80	20.40	160	250	20.80	25.40	200
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	(ii)	Another student B, performed the experiment at room temperature and his graph showing the relationship between volume of $\text{H}_2(\text{g})$ produced at the cathode and time was as shown below. The gradient of the line of best fit gives the volume of hydrogen gas produced per coulomb. Given that the gradient of the line to be $0.125 \text{ cm}^3 \text{ C}^{-1}$, calculate the number of moles of hydrogen gas produced per coulomb. [Molar volume of gas = 24.0 dm^3 at room temperature and pressure] [1]																												
		$\frac{0.125}{24000} = 5.208 \times 10^{-6} \text{ mol C}^{-1}$ [1]																												

		(iii)	Using your answer from (a)(ii) and the half equation for the production of $\text{H}_2(\text{g})$, calculate a numerical value for the Faraday constant (the charge of 1 mole of electrons). [1]
			$1 \div \left(2 \times \frac{0.125}{24000} \right) = 96000 \text{ C mol}^{-1}$ [1]
		(iv)	Using the graph in (a)(ii), draw a line on the graph to show the relationship between volume of $\text{O}_2(\text{g})$ produced at the anode and time in this experiment. [1]
			Straight line from origin to (300, 9.0) [1]
		(v)	Explain why the volume of $\text{O}_2(\text{g})$ measured in the experiment might be less than that shown by your drawn line. Assume that no gas is lost from leaks. [1]
			Oxygen is slightly soluble in water. [1] Or Oxygen gas is assumed to be ideal based on the fact of 24 dm^3 mentioned in the question.
		(b)	A student suggested to replace the platinum electrodes with copper as it would be cheaper in the electrolysis of dilute sulfuric acid. Quoting relevant data from the <i>Data Booklet</i>, suggest what effect, if any, the use of copper electrodes would have on the volume of gas produced at each electrode. [3]
			Cu^{2+}/Cu +0.34 V $\text{O}_2/2\text{H}_2\text{O}$ +1.23 V H^+/H_2 0.00 V Cathode: No effect at cathode. [1] Anode: Copper anode will dissolve as it is a reactive electrode as Cu^{2+}/Cu has a more negative $E^\ominus$ value [1 with data quoted] Less gas produced at anode when copper electrode is used. [1]

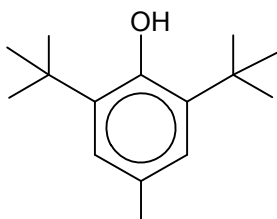
	Reason: <i>not marking for this question but essential for understanding.</i> At the anode: Cu^{2+}/Cu has more negative $E^\ominus$, hence Cu will be oxidised instead of H_2O Anodic reaction when Cu electrode is used: $\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ Anodic reaction when Pt electrode is used: $\text{H}_2\text{O(l)} \rightarrow \frac{1}{2} \text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^-$ At the cathode: Cathodic reaction for Pt and Cu electrode will be the same $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$.
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(c)	Fehling's solution is a copper based chemical reagent used to differentiate between water soluble carbohydrate and ketone functional groups. This test was developed by German chemist Hermann von Fehling in 1849. $\text{RCHO} + 2 \text{Cu}(\text{C}_4\text{H}_4\text{O}_6)_2^{2-} + 5\text{OH}^- \rightarrow \text{RCOO}^- + \text{Cu}_2\text{O} + 4 \text{C}_4\text{H}_4\text{O}_6^{2-} + 3\text{H}_2\text{O}$ Fehling's solution can be used to screen glucose in urine, thus detecting diabetes.  Glucose
(i)	Suggest which carbon can be oxidised by Fehling's solution and draw the skeletal formula of the compound D which is the oxidised product of glucose after it has been acidified by dilute hydrochloric acid. [2]
	Carbon 1 [1] Compound D: [1]
(ii)	Determine the change in oxidation number on the carbon mentioned in (c)(i) with respect to glucose and compound D. [1]

			Oxidation number of carbon 1 on glucose: +1 Oxidation number of carbon 1 on compound D +3 The oxidation number changes from +1 to +3 [1] Or The oxidation number increase by 2 Or +2
		(iii)	Hot concentrated sulfuric acid was added to compound D. Compound E which consists of a 6-atoms ring structure was formed. Suggest the structure of compound E. [1]
			 [1]
			[Total: 13]

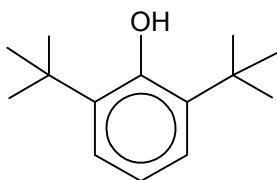
5 This question is about controversial food additives that are Generally Regarded As Safe (GRAS) by the American Food and Drug Administration (FDA), but are suspected to be harmful to humans when consumed.

- (a) Butylated hydroxytoluene (BHT) is useful for its antioxidant properties, to prevent oxidation in fluids (e.g. fuel, oil) and other foodstuffs where free radicals must be controlled.



Butylated hydroxytoluene (BHT)

Describe the mechanism for the synthesis of BHT using bromomethane and 2,6-di-tert-butylphenol shown below.

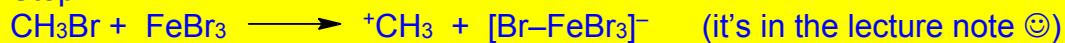


2,6-di-tert-butylphenol

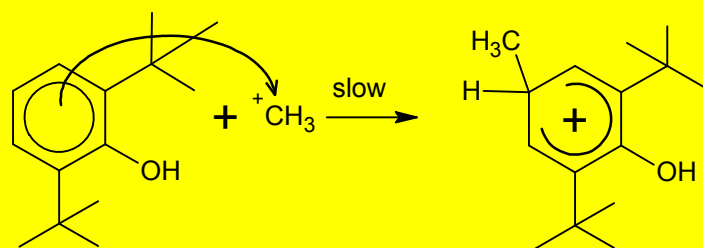
[3]

Electrophilic Substitution

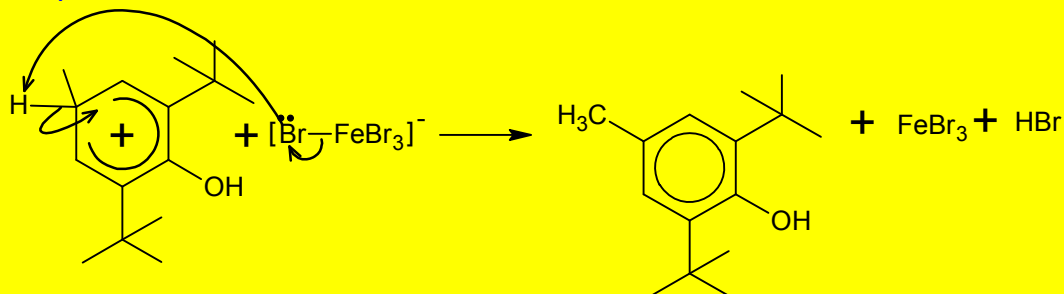
Step 1



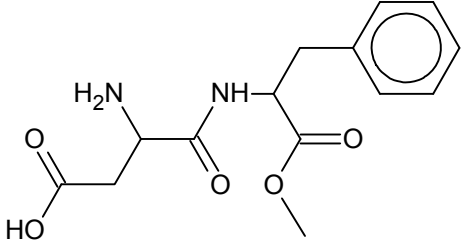
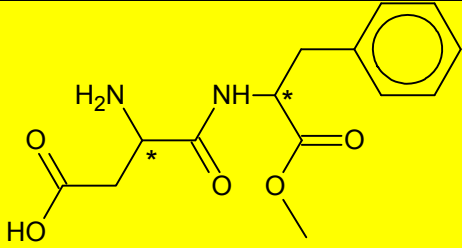
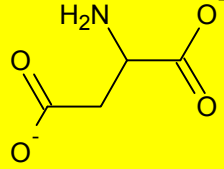
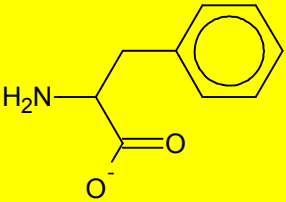
Step 2



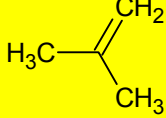
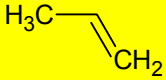
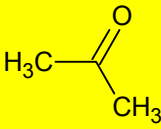
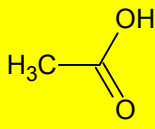
Step 3

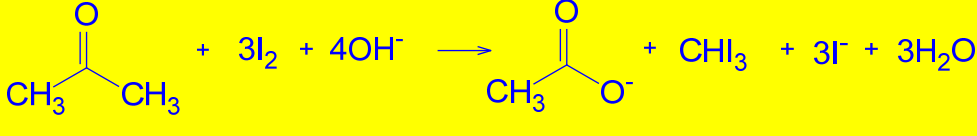
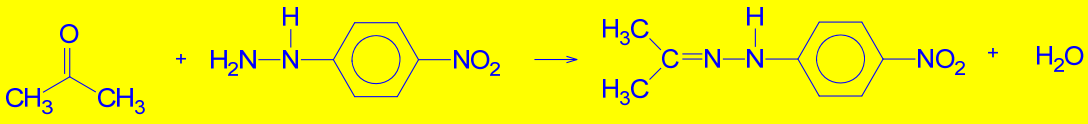
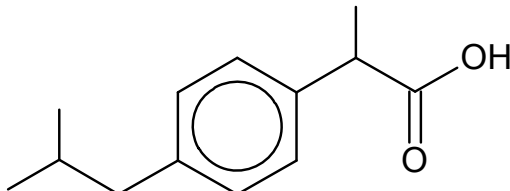
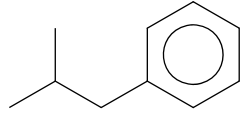


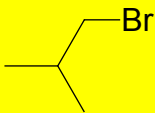
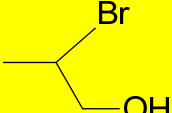
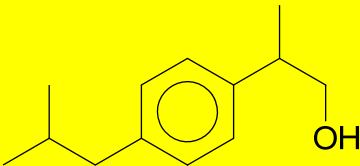
If mis-draw the structure penalise one mark
[3m] each mistake minus 1m

	(a)	(ii)	Propose a simple test-tube reaction to differentiate 2,6-di-tert-butylphenol and BHT. You are to clearly state the observation in your answer. [2]
			Test: Add Br₂ (aq) [1] Observation: Orange Br₂ decolourises in 2,6-di-tert-butylphenol but not in BHT. [1]
	(b)		Aspartame (APM) is an artificial non-carbohydrate sweetener used as a sugar substitute in some foods and beverages, which is especially useful for diabetic patients.  Aspartame (APM)
		(i)	Label all chiral carbons with an (*) in the diagram above and calculate the total number of possible stereoisomers of APM. [2]
			 [1] $2^2 = 4$ isomers [1]
		(ii)	Draw all organic products formed when APM is reacted with a hot aqueous alkaline solution. [3]
			  H₃C—OH [1] for each correct structure

		(c)	Potassium bromate, KBrO_3, is typically used as a flour improver. It is a very strong oxidising agent, and allows for the baking of extremely white and fluffy bread, and it has the following half equation. $2\text{BrO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightleftharpoons \text{Br}_2 + 6\text{H}_2\text{O} \quad E^\ominus = +1.50 \text{ V}$
		(i)	Suggest reagent and conditions for the oxidation of ethene to carbon dioxide.
			KMnO_4 , H_2SO_4 (aq), Heat [1]
		(ii)	Hot acidified bromate is also able to oxidise ethene into carbon dioxide. With reference to the <i>Data Booklet</i> , explain. [1]
			$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad E = +1.52 \text{ V}$ Since the <u>reduction potential of bromate is similar to that of manganate(VII)</u> , bromate would be a strong enough oxidising agent to oxidise ethene to carbon dioxide. [1]
		(iii)	Hence, suggest how the yield of the carbon dioxide produced would be affected due to the use of bromate as an oxidising agent. [1]
			The yield of CO_2 produced would be lower than expected as Br_2 would be produced which would then react with some ethene reactant present. [1]
			[Total: 13]

6	Compound G and H have molecular formula of C_4H_8 and C_3H_6 respectively, and do not exhibit cis-trans isomerism. Both G and H can decolourise hot potassium manganate(VII) to give compounds J and K respectively, together with a gas that produces a white precipitate when bubbled through calcium hydroxide. J produces an orange precipitate when warm with 2,4–dinitrophenylhydrazine, and K produces effervescence when sodium hydrogencarbonate is added. J decolourises alkaline aqueous iodine but not K.
(a)	Draw the structures of G , H , J , and K . [4]
	 G  H  J  K [1] for each correct structure Statements for your information (not required by question) ✓ Both G and H has molecular formula with general formula C_nH_{2n} • G and H are <u>alkenes</u> ✓ Both G and H decolourises hot $KMnO_4$ to form J and K and CO_2 • <u>oxidation</u> • G and H are <u>terminal alkenes</u> ✓ J produces orange ppt with 2,4–DNPH • <u>condensation</u> • J is a <u>ketone</u> ✓ J decolourises alkaline I_2 • <u>Oxidation</u> • J has $\begin{array}{c} O \\ \\ R-C-CH_3 \end{array}$ ✓ K produces gas with $NaHCO_3$ • <u>neutralisation</u> • K is a <u>carboxylic acid</u>

	(b) Construct a balanced chemical equation on how J reacts with - alkaline aqueous iodine, 2,4-DNPH. [2]
	 [1]  [1]
	(c) Ibuprofen is an oral or intravenous medication that is used for treating pain, fever, and inflammation. It typically begins working within an hour and it has the following structure.  Ibuprofen Compounds G and H from (a) are both prerequisites for the synthesis of Ibuprofen. The reaction schema to obtain ibuprofen is described below. (1) HBr gas is introduced to G to form L. (2) L is reacted with benzene in the presence of anhydrous FeBr₃. Compound M is obtained.  Compound M (3) Aqueous Br₂ is added to H to form N. (4) N and M are reacted in the presence of anhydrous FeBr₃. Compound P with molecular formula C₁₃H₂₀O is produced. (5) Hot acidified potassium dichromate(VI) is then added to P to produce ibuprofen. Using the structure of ibuprofen and the information provided, draw the structures of L, N and P. [3]

		L:  [1] (only allow 1°) N:  [1] p:  [1]
		[Total: 9]

END OF PAPER